

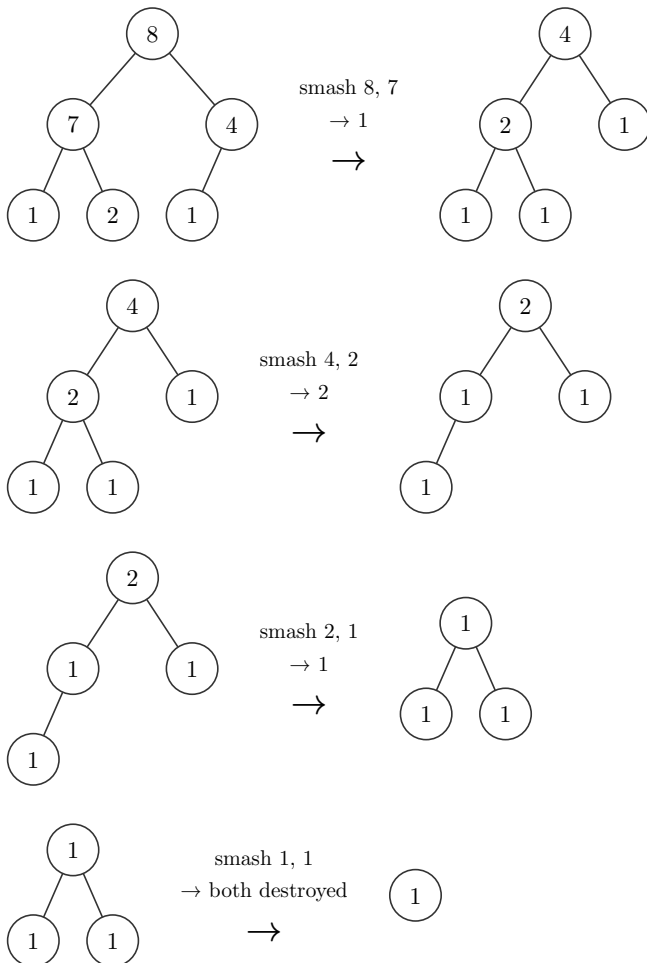
Last Stone Weight

LeetCode 1046 · Easy

Smash the two heaviest stones together, repeatedly, until at most one is left. If they're equal both are destroyed; otherwise the difference goes back in. A max-heap hands you the two heaviest in $O(\log n)$ each round instead of a full rescan.

```
1 import heapq
2
3 def lastStoneWeight(stones):
4     heap = [-s for s in stones]
5     heapq.heapify(heap)
6     while len(heap) > 1:
7         y, x = -heapq.heappop(heap), -heapq.heappop(heap)
8         if y != x:
9             heapq.heappush(heap, -(y - x))
10    return -heap[0] if heap else 0
```

 Python



One stone remains, weight 1 — matching `lastStoneWeight([2, 7, 4, 1, 8, 1]) == 1`.